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AnalystLab Africa - Data Science Internship Programme

Week 5 Assignment

Week 5 Project Summary

**Improving Patient Appointment Attendance and Healthcare Support Using Data and
AI**

HealthConnect Clinic Experience Lab

Name: Wilson Moses

Internship: Data Science

Batch: D

Submission Date: Sun. 06.09.2026

HealthConnect | Week 5 Project Summary

Wilson Moses | AnalystLab Africa | Data Science Track | Batch D

What I planned and completed

I planned to move from the Week 4 problem definition into preparation, feature engineering and an initial predictive baseline. I completed an executable notebook, source-quality checks, training-only visual exploration, four engineered features, a chronological split, pipeline-based preprocessing, two classifiers and held-out evaluation. I preserved the original data and exported the cohort, metrics, predictions and decision evidence.

Key results

Measure	Result
Source / binary cohort	5,000 / 4,737 appointments
Training / test / purged	3,682 / 799 / 256
Logistic accuracy / ROC-AUC	60.83% / 0.6678
Logistic precision / recall	58.85% / 63.57%
Flagged / actual no-shows caught	418 / 246
False positives / missed no-shows	172 / 141

The dummy flags every appointment and has higher recall and F1. Logistic regression provides better accuracy and ranking with fewer flags, but this is a limited educational baseline, not an established operational solution. No clinic improvement or real-world generalisation has been demonstrated.

Important decisions and changes from Week 4

I refined scoring to booking time and excluded reminders and waiting time because their timing is unknown. I retained the cancellation exclusion, preserved Sunday conflicts for investigation and used supplied history only under an explicit pre-booking assumption. I strengthened the time split by purging appointments whose outcomes cross the cutoff, and confined model-directed exploration to training data.

Challenges, remaining work and Week 6

The main challenges were missing distance, source date-format differences, 737 Sunday appointments and inconsistent ages across repeated patient IDs. Parsing is corrected and distance imputation is training-only; scheduling conflicts and historical-data reliability remain unresolved. There are 579 shared patient IDs across partitions, so unseen-patient generalisation is not established.

Cross-track collaboration status

Pending: I prepared an ML Engineering handoff but have not documented a verified exchange with another track. I must complete the interaction and record the person/track, exchanged material, relevance and resulting improvement. I will not describe this prepared document as completed collaboration.

Proposed Week 6 focus

- Confirm history provenance and feature availability at booking.
- Use training-only chronological validation for no-history, Sunday and simple-tree comparisons.
- Review subgroup errors and probability calibration with adequate support.
- Agree follow-up capacity and choose a validation-based threshold.
- Revisit future cancellations and reserve fresh evaluation data before further claims.

Submission checks still requiring my action

I must review and understand the AI-assisted draft, complete the real collaboration record, merge the supplied README guidance with my existing branded repository, upload the files to the designated Drive folder, confirm mentor access and submit the official form. None of those external actions has been performed by this package.

Conclusion

Week 5 establishes a reproducible baseline rather than a finished clinic system. The most important progress is the link between my Week 4 decisions, the actual preparation steps and an honest interpretation of model errors. The results now give me a defensible starting point for Week 6.